Answers · Term 2

TERM 2

Teacher copy. Answers were calculated in code and checked. Circumference and Pythagoras answers use the π and √ keys; where π ≈ 3.14 gives a different 1 d.p. answer it is listed as an alternative. Reasoning answers are model responses: accept any correct explanation.

WK 1	Q1	Q2	Q3	Q4	Q5
S1	$x = 7$ m	5% profit	9	The square: rectangle 30 m², square 36 m².	2 L
S2	$165	9^7	88 cm²	$\frac{6}{7}$	$a = 3$
S3	$x = 1$ cm	120 students	(−1, 2)	$x = 60$	3.47

WK 2	Q1	Q2	Q3	Q4	Q5
S1	7^2	$x = 5.8$ cm	$x = 76$	5	42 m
S2	10 and 11 (because 100 < 111 < 121)	$7	$x = 67$	No: P(yellow) = $\frac{1}{4}$, which is less than $\frac{1}{2}$.	51.9
S3	$x = 120$ m	50%	$k = 17$	$x = 54$	60%

WK 3	Q1	Q2	Q3	Q4	Q5
S1	Yes: $6^2 + 8^2 = 100 = 10^2$	4^{12}	$0.\dot{6}$	$x = 49$	33
S2	$x = 30$ cm	$396	$x = 122$	$\frac{21}{50}$ (0.42)	$k = -1$
S3	$x = 6$ cm	72 km	$6x$	4	$x = 77$ (angle sum of a triangle is 180°)

WK 4	Q1	Q2	Q3	Q4	Q5
S1	Grace only multiplied the first term; both terms must be multiplied by 2: $2m + 4$.	$x = 130$ cm	300 mm²	116	10
S2	$9(2y + 3)$	5	$x = 45$	14 cm²	75%
S3	$5m + 26$	Yes: $10^2 + 24^2 = 676 = 26^2$	2	$x = 67$	5

WK 5	Q1	Q2	Q3	Q4	Q5
S1	$6x$	25% profit	$x = 140$	$\frac{3}{5}$	77 m²
S2	4^5	$x = 9$ cm	Yes: substituting $m = 11$ makes both sides equal.	10	$x = 59$
S3	$2m^2 + 9m$	5^4	96 m³	$2 \times 2 \times 13$	$x = 67$

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WK 6	Q1	Q2	Q3	Q4	Q5
S1	$2^4 \times 5$	$x = 100$ cm	$x = 0$: $y = 1$, $x = 2$: $y = 3$	−18	$x = 63$
S2	96 kg	1	8	200 cm^3	$x = 132$
S3	$12m - 27$	$x = 8.1$ cm	$x = 63$	85%	Mia multiplied 11 × 2; 11^2 means 11 × 11 = 121.

WK 7	Q1	Q2	Q3	Q4	Q5
S1	Yes: both are $4 (5% of 80 = 80% of 5 = 5 × 80 ÷ 100).	No: $6^2 + 7^2 = 85$, but $9^2 = 81$	1	8 students	$4p + 8$
S2	$8(4m + 3)$	4^{16}	$x = 103$	37.7 cm	$192
S3	$3n + 22$	9^9	12	$x = 59$	46 m

WK 8	Q1	Q2	Q3	Q4	Q5
S1	$-12y - 24$	$x = 24$ m	$\frac{11}{60}$ (0.183)	$x = 61$	3.2
S2	160 students	$3(4x + 1)$	$x = 36$	4	The rectangle: rectangle 40 m^2, square 36 m^2.
S3	$x = 26$ cm	$8x^2$	$\frac{1}{7}$	104 cm^2	$\frac{2}{7}$

WK 9	Q1	Q2	Q3	Q4	Q5
S1	$121	$x = 3$	85%	225 m^2	27
S2	$l = 4$	9 and 10 (because 81 < 93 < 100)	$x = 113$	0.26	133 cm
S3	$a = 6$ or $a = -6$	$-2x + 10$	$\frac{3}{20}$	The mean (9) is larger than the median (6) because the outlier 23 pulls the mean up.	250 mL

WK 10	Q1	Q2	Q3	Q4	Q5
S1	$5n - 4 = 36$; the number is 8	$2(2m + 7)$	0.7	$x = 160$ (angles at a point add to 360°)	99 m^2
S2	$x = 2.5$ (= $2\frac{1}{2}$)	No: $2^2 + 3^2 = 13$, but $4^2 = 16$	10	$x = 31$	1
S3	$x = 3$	$4n + 21$	96 cm^2	$x = 115$	44